

Solving Two-Step Equations

MA.7.AR.2.2

I can solve a two-step equation and name the property of equality at each step.

The most content in one period all unit.

An equal sign is a claim of balance, not an instruction to compute.

Warm-Up

On your own. Four minutes.

Spaced retrieval

1. Last lesson: write $0.454545\dots$ as a fraction.
2. Last week: write $\frac{5}{8}$ as a decimal and as a percent.
3. Prior grade: what is $7 - (-3)$?
4. Prior grade: what is $(-4)(-6)$?

Notes

I. An equal sign is a claim of balance

A. equation — two expressions with the same value

B. **The equal sign is not an instruction to compute. It is a claim.**

C. solution — the value that makes the equation true

Notes

II. The properties of equality

Property	If $a = b$, then...	Example, from $7 = 3 + 4$
Symmetric	$b = a$	$3 + 4 = 7$
Addition	$a + c = b + c$	$7 + 2 = 3 + 4 + 2$
Subtraction	$a - c = b - c$	$7 - 2 = 3 + 4 - 2$
Multiplication	$a \cdot c = b \cdot c$	$7 \times 2 = (3 + 4) \times 2$
Division, $c \neq 0$	$a \div c = b \div c$	$7 \div 2 = (3 + 4) \div 2$

Notes

III. Undo the outside layer first

- A. **Undo the addition or subtraction first.**
- B. **Then undo the multiplication or division.**
- C. Name the property you used at each step.

Notes

IV. Check by substituting into the ORIGINAL

- A. **Put your answer back into the equation you were given.**
- B. Not into a line of your own working — that only proves you can copy.
- C. **Work both sides separately and see whether they match.**

Example 1

Which layer comes off first?

$$5x - 9.6 = 21.4$$

Answer out loud, and name the property.

Example 1

Two steps, two properties.

$$5x - 9.6 + 9.6 = 21.4 + 9.6 \quad \text{addition property of equality}$$

$$5x = 31$$

$$5x \div 5 = 31 \div 5 \quad \text{division property of equality}$$

$$x = 6.2$$

Example 1

Check it.

$$5(6.2) - 9.6 = 31 - 9.6 = 21.4 \quad \checkmark$$

Example 2

The variable is on the right.

$$-6.7 = n/4 + 1.3$$

Does that change anything?

Example 2

Same two steps.

$$-6.7 - 1.3 = n/4 + 1.3 - 1.3 \quad \text{subtraction property of equality}$$

$$-8 = n/4$$

$$-8 \times 4 = (n/4) \times 4 \quad \text{multiplication property of equality}$$

$$-32 = n, \text{ so } n = -32 \quad \text{symmetric property}$$

Example 2

Check it.

$$-32 \div 4 + 1.3 = -8 + 1.3 = -6.7 \quad \checkmark$$

Example 3

Start from $5.5 + 8 = 13.5$. Which of these stays true?

What we do	Result	Still true?
Add 23 to the left, 14 to the right	$36.5 = 27.5$	
Multiply the left by 3, the right by -1	$40.5 = -13.5$	
Add 5 to both sides	$18.5 = 18.5$	
Divide both sides by 0		

Example 3

Three answers, and one that is not an answer.

What we do	Result	Still true?
Add 23 to the left, 14 to the right	$36.5 = 27.5$	No
Multiply the left by 3, the right by -1	$40.5 = -13.5$	No
Add 5 to both sides	$18.5 = 18.5$	Yes
Divide both sides by 0	no equation at all	—

Example 3

Why only one property carries a condition.

How many times does 0 go into 13.5?

- A. NO answer works. Whatever you put, 0 times it is 0 — and 0 is not 13.5.
- B. **5.5 + 8 divided by 0 is not a big number and it is not zero. There is nothing to write.**
- C. That is the whole reason the division property says $c \neq 0$, and it says so all through Algebra 1.

Whiteboards • Question 1 of 9

Boards up on three.

$$2x + 7 = 19$$

Solve it.

One value. Then check it in the original.

Whiteboards • Question 1 of 9

Answer.

$$2x + 7 = 19$$

subtract 7, then divide by 2

Answer: $x = 6$

Whiteboards • Question 2 of 9

Boards up on three.

$$2x - 7 = 19$$

Solve it.

One value. Then check it in the original.

Whiteboards • Question 2 of 9

Answer.

$$2x - 7 = 19$$

ADD 7, then divide by 2

Answer: $x = 13$

Whiteboards • Question 3 of 9

Boards up on three.

$$-2x + 7 = 19$$

Solve it.

One value. Then check it in the original.

Whiteboards • Question 3 of 9

Answer.

$$-2x + 7 = 19$$

subtract 7, then divide by -2

$$\textbf{Answer: } x = -6$$

Whiteboards • Question 4 of 9

Boards up on three.

$$x \div 2 + 7 = 19$$

Solve it.

One value. Then check it in the original.

Whiteboards • Question 4 of 9

Answer.

$$x \div 2 + 7 = 19$$

subtract 7, then MULTIPLY by 2

Answer: $x = 24$

Whiteboards • Question 5 of 9

Boards up on three.

$$19 = x \div 2 + 7$$

Solve it.

One value. Then check it in the original.

Whiteboards • Question 5 of 9

Answer.

$$19 = x \div 2 + 7$$

the sides are swapped; nothing else is

Answer: $x = 24$

Whiteboards • Question 6 of 9

Boards up on three.

$$7 + x \div 2 = 19$$

Solve it.

One value. Then check it in the original.

Whiteboards • Question 6 of 9

Answer.

$$7 + x \div 2 = 19$$

the terms are swapped; still nothing else

Answer: $x = 24$

Whiteboards • Question 7 of 9

Boards up on three.

$$\frac{2}{3}x + 7 = 19$$

Solve it.

One value. Then check it in the original.

Whiteboards • Question 7 of 9

Answer.

$$\frac{2}{3}x + 7 = 19$$

subtract 7, then multiply by $\frac{3}{2}$

Answer: $x = 18$

Whiteboards • Question 8 of 9

Boards up on three.

$$0.5x + 7 = 19$$

Solve it.

One value. Then check it in the original.

Whiteboards • Question 8 of 9

Answer.

$$0.5x + 7 = 19$$

0.5x means $x \div 2$

Answer: $x = 24$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Maya solved $-2x + 7 = 19$ and got $x = 6$. Check her answer, then give the right one.

Write your answer in sentences.

This one is written work. Show the substitution.

Whiteboards • Question 9 of 9

Answer.

Maya solved $-2x + 7 = 19$ and got $x = 6$. Check her answer, then give the right one.

$$-2(6) + 7 = -12 + 7 = -5, \text{ and } -5 \text{ is not } 19$$

Answer: $x = -6$

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Whatever you do not finish is tonight's practice.
4. The skills for today are posted in Google Classroom.